Metrics & Evaluation

Applied Machine Learning — Session 2, Chapter 3

The Metric Defines Success

> *A model can only be as good as your definition of "good."*

Example: 99% of patients are healthy.

→ Model that always predicts "healthy" = 99% accurate.

→ But it catches 0% of sick patients.

Choosing the wrong metric = optimizing for the wrong thing.

Regression Metrics

Metric	Formula	Units	Best
MAE	$(1/n)\Sigma\|y_i - \hat{y}_i\|$	Same as y	Low
MSE	$(1/n)\Sigma(y_i - \hat{y}_i)^2$	y^2	Low
RMSE	$\sqrt{MSE}$	Same as y	Low
R^2	1 - SS_res/SS_tot	Unitless	→ 1

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

RMSE vs MAE: RMSE penalizes large errors more (because of squaring).

Use RMSE when big mistakes are costly.

R^2 Score

"How much of the variance in y does our model explain?"

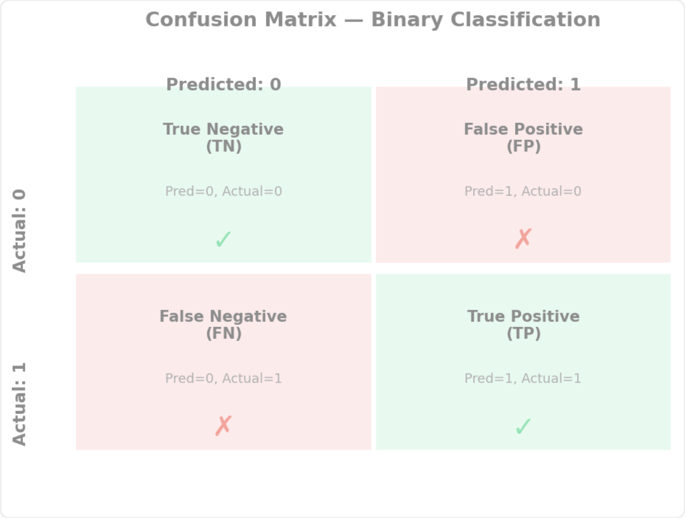
```
R² = 1.0  → perfect predictions
R² = 0.0  → no better than predicting the mean
R² < 0.0  → worse than predicting the mean (very bad!)
```

```
r2 = r2_score(y_test, y_pred)
print(f'R² = {r2:.3f}')
```

A good R^2 depends on the domain:

- Finance: R^2=0.1 can be impressive
- Physics: R^2=0.99 is expected

The Confusion Matrix




```python
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
cm = confusion_matrix(y_test, y_pred)
ConfusionMatrixDisplay(cm).plot()
```

Reading the Confusion Matrix

```
Spam filter example:
                Predicted: Ham  |  Predicted: Spam
Actual: Ham        TN=90        |    FP=5    ← "lost" emails ⚠️
Actual: Spam       FN=3         |    TP=2    ← caught spam ✅
```

- **FP (False Positive):** Ham classified as spam → Important email goes to spam folder!
- **FN (False Negative):** Spam not caught → Gets through to inbox

Different errors have different costs — always think about consequences.

Precision, Recall, F1

Precision: When I predict positive, how often am I right?

```
Precision = TP / (TP + FP)
```

Recall: Of all actual positives, how many did I find?

```
Recall = TP / (TP + FN)
```

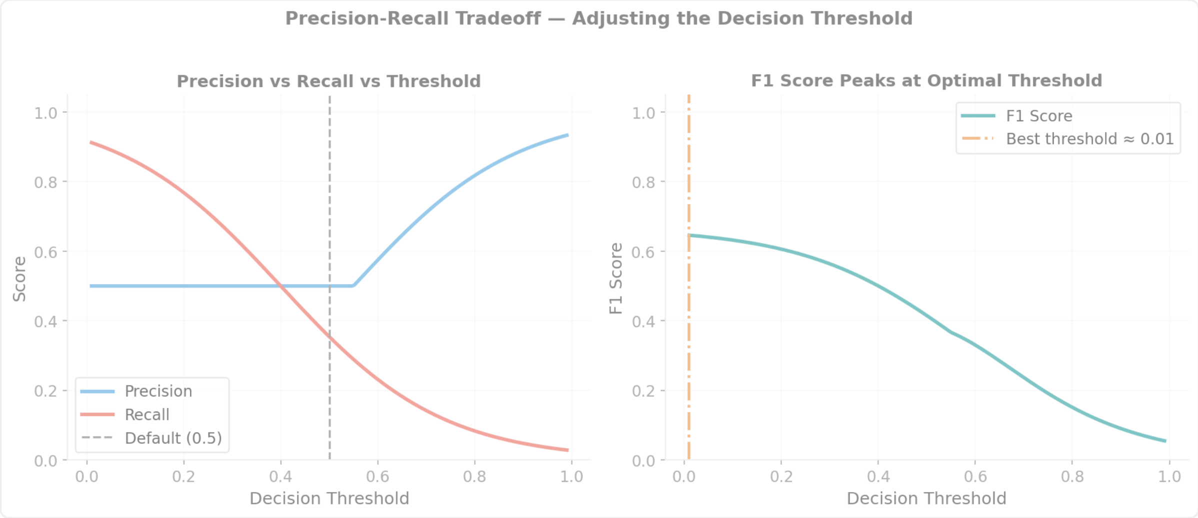
F1 Score: Balanced summary

```
F1 = 2 × Precision × Recall / (Precision + Recall)
```

When to Use Which

Metric	Optimize when...	Example
Precision	FP is costly	Spam filter (don't lose good email)
Recall	FN is costly	Disease screening (don't miss sick patients)
F1	Both matter	General classification
Accuracy	Classes are balanced	Digit recognition (10 balanced classes)

The Precision-Recall Trade-off



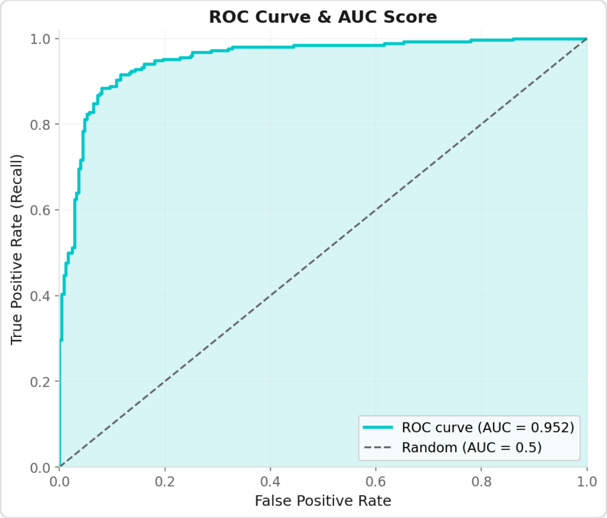
Adjusting the decision threshold:

```
threshold = 0.3 → more positives → higher recall, lower precision
threshold = 0.7 → fewer positives → lower recall, higher precision
```

ROC Curve & AUC

ROC = plot of True Positive Rate vs False Positive Rate

across all possible decision thresholds.



- **AUC = 1.0:** Perfect
- **AUC = 0.5:** Random guessing (the diagonal)

Cross-Validation for Model Comparison

```python
from sklearn.model_selection import cross_val_score

models = {
    'Logistic': LogisticRegression(),
    'KNN': KNeighborsClassifier(5),
    'Random Forest': RandomForestClassifier(100),
}

for name, model in models.items():
    scores = cross_val_score(model, X, y, cv=5, scoring='f1')
    print(f'{name}: {scores.mean():.3f} ± {scores.std():.3f}')
```

Look at both mean AND standard deviation.

Now: Exercises!

→ Open `03-exercises/ch06_metrics_exercises.ipynb`

You will:

- Compute regression metrics for a diabetes progression model
- Build and read a confusion matrix
- Calculate precision, recall, F1
- Plot an ROC curve

~10 minutes

Key Takeaways

- Accuracy is misleading for imbalanced data
- Choose metrics that match your real-world cost of errors
- Confusion matrix: understand TP, TN, FP, FN
- Precision vs Recall trade-off — adjust threshold
- AUC: overall model quality, threshold-independent
- Cross-validation: always use it for model comparison

Next: Session 3

Unsupervised Learning

"What if we don't have labels? What can we still learn?"